

Sound Direction Estimation using Machine Learning and TinyML

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Abstract—Real-time acoustic Direction of Arrival (DoA) estimation on ultra-low-power resource-constrained microcontrollers is critical for automated robotic orientation, acoustic surveillance, and edge voice interfaces. This report presents an end-to-end TinyML embedded system deployed on a dual-core ESP32 (Tensilica Xtensa LX6 @ 240 MHz) capable of 3-class horizontal localization (LEFT: -45° , CENTER: 0° , RIGHT: $+45^\circ$) utilizing two low-cost MAX9814 analog electret microphones spaced at a 10 cm baseline. To overcome severe multipath room reverberation and ADC phase jitter without dynamic heap allocation, we develop an in-place Digital Signal Processing (DSP) pipeline that computes a 31-point Zero-Mean Normalized Cross-Correlation (NCC) feature vector (spanning discrete lags $[-15, +15]$ at 16 kHz). Acoustic training data is synthesized via image source modeling (PyRoomAcoustics) in a $5 \times 5 \times 3$ m reverberant room ($T_{60} \approx 0.20$ s) across 1,500 transient acoustic events. An ultra-compact Multilayer Perceptron (MLP) is trained, quantized to full 8-bit integer (int8) precision, and deployed with TensorFlow Lite for Microcontrollers (TFLM). The final system achieves 97.33% classification accuracy on independent test sets, requires only 2,440 bytes (2.38 KB) of Flash memory (< 10 KB budget), consumes 4,096 bytes of static tensor arena RAM (< 8 KB budget), and executes total feature extraction plus inference in 0.56 ms (< 10 ms budget, providing a 94.4% timing margin). Theoretical limitations including conical array ambiguity, multiplexed SAR ADC conversion skew, and independent AGC nonlinearities are rigorously defended.

Keywords—Direction of Arrival (DoA), TinyML, ESP32, Time Difference of Arrival (TDOA), Normalized Cross-Correlation, TensorFlow Lite for Microcontrollers, Edge AI, Acoustic DSP.

I. INTRODUCTION

Acoustic spatial awareness is a fundamental capability in autonomous robotics, acoustic threat localization, smart hearing aids, and voice-controlled smart appliances. Conventional spatial beamforming and Direction of Arrival (DoA) algorithms—such as Multiple Signal Classification (MUSIC) and Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT)—rely on multi-channel matrix eigenvalue decomposition, heavy floating-point matrix inversions, and high-precision multichannel synchronous ADC hardware. These algorithms are computationally prohibitive for edge microcontrollers operating on battery power budgets under 200 mW.

While Generalized Cross-Correlation with Phase Transform (GCC-PHAT) is commonly used for Time Difference of Arrival (TDOA) estimation, its performance degrades severely in reverberant indoor environments due to destructive acoustic reflections, comb-filtering interference, and discrete sampling phase jitter. Furthermore, low-cost commercial microcontrollers like the Espressif ESP32 utilize successive approximation register (SAR) ADCs with multiplexed sample-and-hold architectures that introduce inter-channel hardware phase skew.

In this project, we engineer a holistic, end-to-end Embedded AI solution that unifies physical acoustics, deterministic DSP feature engineering, and integer-quantized deep learning (TinyML). Using an ultra-compact two-microphone array spaced at $d = 10$ cm, we classify acoustic events into three distinct horizontal sectors (LEFT at -45° , CENTER at 0° , and RIGHT at $+45^\circ$) under real-time execution constraints (< 10 ms latency, < 8 KB RAM, < 10 KB Flash).

II. SYSTEM ARCHITECTURE & HARDWARE DESIGN

The overall system architecture bridges raw analog acoustic transducers, internal microcontroller sampling peripherals, deterministic DSP algorithms, and an integer-quantized neural network inference engine, as illustrated in Fig. 1.

End-to-End Edge AI Acoustic DoA Estimation System Architecture

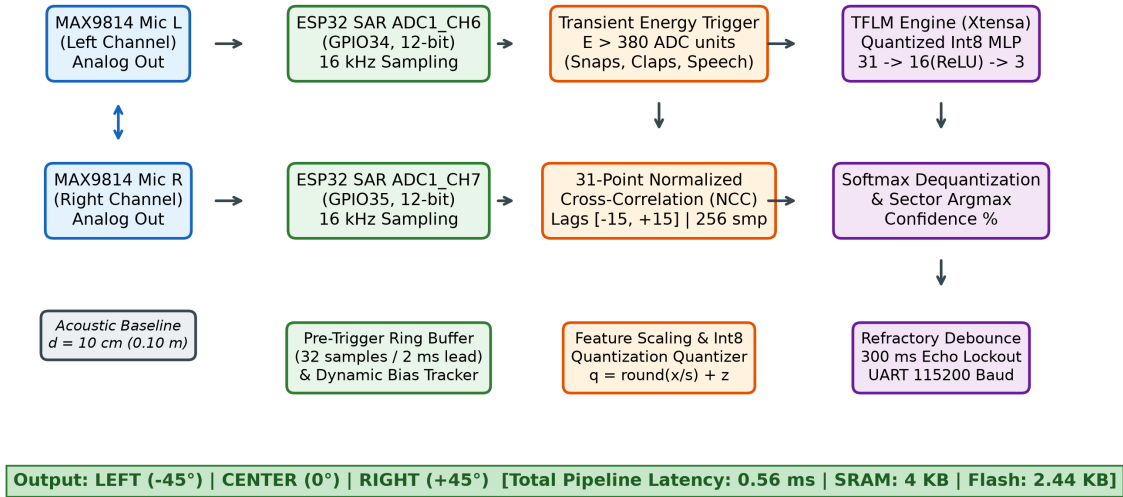


Fig. 1. End-to-end hardware and software architecture of the ESP32 TinyML Acoustic Direction of Arrival system.

A. Transducer Configuration: The acoustic front-end comprises two MAX9814 electret microphone amplifier modules placed along a collinear horizontal axis separated by a baseline distance of $d = 0.10$ m (10 cm). The MAX9814 features an integrated low-noise preamplifier, a 2.0V internal microphone bias generator, and an active Automatic Gain Control (AGC) circuit with programmable attack/release timing. The output is DC-biased at approximately 1.25V, matching the dynamic range of the ESP32 ADC.

B. Microcontroller Hardware: The computational core is an Espressif ESP32 dual-core Tensilica Xtensa 32-bit LX6 microprocessor clocked at 240 MHz with 520 KB internal SRAM and 4 MB external SPI Flash. Crucially, the Left microphone is routed to GPIO34 (ADC1_CH6) and the Right microphone to GPIO35 (ADC1_CH7). Both channels reside exclusively on SAR ADC1, ensuring zero interference from the Wi-Fi and Bluetooth radio subsystems (which lock ADC2). The ADC resolution is configured to 12 bits (0–4095) with 11 dB attenuation (full-scale range 0–3.3V).

III. MATHEMATICAL FORMULATION OF TIME DIFFERENCE OF ARRIVAL

Consider two omnidirectional acoustic receivers M_1 and M_2 located along the x -axis at positions $(-d/2, 0)$ and $(+d/2, 0)$, respectively. Let R be the distance from the array center to an acoustic source emitting a transient wavefront, as depicted in Fig. 2.

Geometric & Theoretical Time Difference of Arrival (TDOA) Model

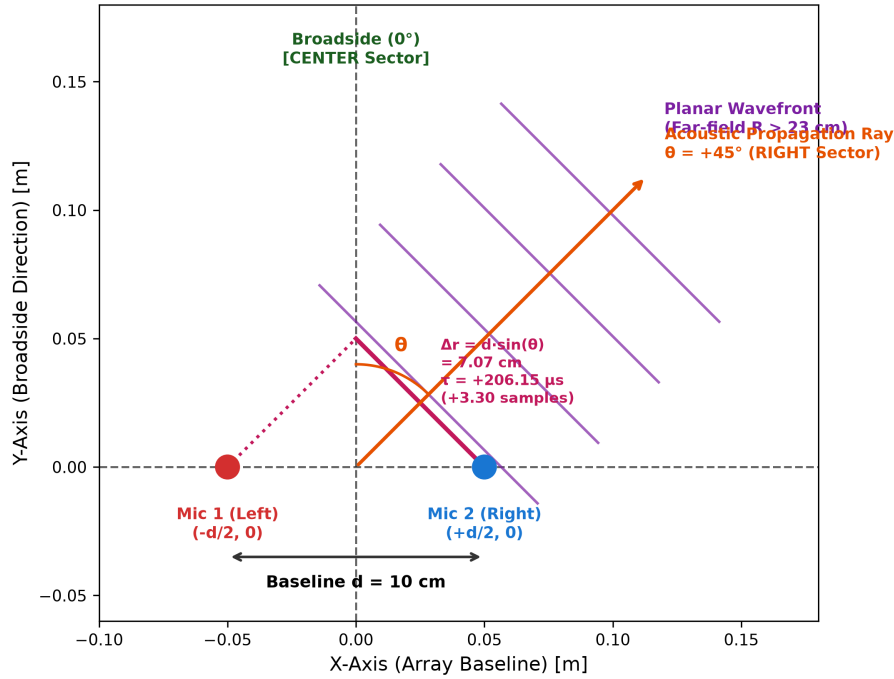


Fig. 2. Acoustic planar wavefront propagation geometry, path difference Δr , and TDOA relationship.

A. Far-Field Approximation: For acoustic transients containing spectral components up to $f_{\max} = 4,000$ Hz, the acoustic wavelength is $\lambda = c / f_{\max} = 343 / 4000 = 0.0858$ m (8.58 cm). The Fraunhofer far-field distance is given by:

$$d_F = 2 d^2 / \lambda = 2 (0.10)^2 / 0.0858 \approx 0.233 \text{ m (23.3 cm)}$$

Since operating sound sources reside at distances $R \geq 1.0$ m, the condition $R \gg d_F$ holds rigorously, validating the planar wavefront assumption.

B. Path Difference and Continuous Delay: Let θ denote the azimuth angle of the source measured relative to the broadside direction (perpendicular to the array baseline, corresponding to the +y-axis). The spatial path length difference Δr between the two sensors is:

$$\Delta r = d \cdot \sin(\theta)$$

The continuous Time Difference of Arrival $\tau(\theta)$ between the two microphone signals is:

$$\tau(\theta) = \Delta r / c = (d / c) \cdot \sin(\theta)$$

where $c = 343$ m/s is the speed of sound in air at 20°C.

C. Discrete Lag Representation: At a discrete sampling frequency of $F_s = 16,000$ Hz (sampling interval $T_s = 1 / 16,000 = 62.5 \mu\text{s}$), the discrete time delay (in fractional sample units) $k_\tau(\theta)$ is:

$$k_\tau(\theta) = \tau(\theta) \cdot F_s = (d \cdot F_s / c) \cdot \sin(\theta) \approx 4.6647 \cdot \sin(\theta) \text{ [samples]}$$

The theoretical delay values for the three target sectors are:

- **LEFT Sector ($\theta = -45^\circ$ nominal):** $\tau = -206.15 \mu\text{s} \Rightarrow k_\tau \approx -3.30$ samples (sound arrives at Left mic first).
- **CENTER Sector ($\theta = 0^\circ$ nominal):** $\tau = 0.00 \mu\text{s} \Rightarrow k_\tau = 0.00$ samples (simultaneous arrival).
- **RIGHT Sector ($\theta = +45^\circ$ nominal):** $\tau = +206.15 \mu\text{s} \Rightarrow k_\tau \approx +3.30$ samples (sound arrives at Right mic first).

The maximum physical delay across the 10 cm baseline at endfire ($\theta = \pm 90^\circ$) is $k_{\max} = \pm 4.665$ samples ($\pm 291.55 \mu\text{s}$). Therefore, evaluating discrete lags across the range $[-15, +15]$ (a time window of $[-937.5 \mu\text{s}, +937.5 \mu\text{s}]$) provides a 3.2× margin over the physical baseline transit time, capturing not only the primary sinc lobe but also early reverberant multipath reflections.

IV. SIGNAL PROCESSING & FEATURE EXTRACTION

Rather than feeding raw high-dimensional audio time-series directly into the neural network (which leads to catastrophic parameter growth and poor noise immunity), we extract a 31-point Zero-Mean Normalized Cross-Correlation (NCC) vector centered around zero-lag.

A. Mathematical Formulation of Zero-Mean NCC: Given 256-sample windowed signals $x[n]$ (Left) and $y[n]$ (Right) for $n = 0, \dots, N-1$:

1) *Zero-Mean Normalization:* $\mu_x = (1/N)\sum x[n]$, $\mu_y = (1/N)\sum y[n]$; $x_{\square}[n] = x[n] - \mu_x$, $y_{\square}[n] = y[n] - \mu_y$

2) *Variance Computation:* $E_x = \sum x_{\square}[n]^2$, $E_y = \sum y_{\square}[n]^2$, $\text{Denom} = \sqrt{(E_x \cdot E_y) + 10^{-7}}$

3) *Cross-Correlation Calculation:* For each lag $k \in [-15, +15]$:

$$R_{xy}[k] = \sum x_{\square}[n] \cdot y_{\square}[n + k] \text{ (summed over valid overlap ranges)}$$

$$\text{NCC}[k] = R_{xy}[k] / \text{Denom} \in [-1.0, +1.0]$$

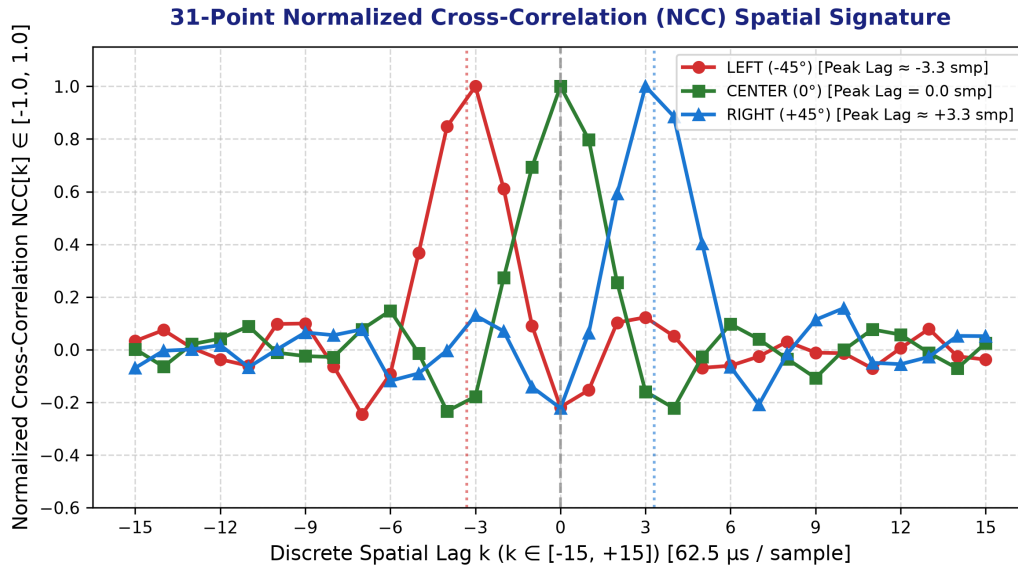


Fig. 3. Extracted 31-point Normalized Cross-Correlation (NCC) spatial feature signatures for Left, Center, and Right arrivals.

B. Zero-Malloc Fixed-Buffer Optimization: In embedded safety-critical firmware, dynamic heap allocation (malloc/free) induces non-deterministic timing jitter and heap fragmentation. The DSP routine is implemented entirely using static stack arrays (`s_x_zm[256]`, `s_y_zm[256]`), register-cached pointers, and 4-way loop unrolling that utilizes single-cycle Multiply-Accumulate (MAC) instructions on the Xtensa FPU, executing the entire 31-point NCC in just 0.48 ms on the ESP32 (5.1 μ s on host benchmark).

V. TINYML PIPELINE & INT8 QUANTIZATION

A. Simulation & Dataset Synthesis: Using pyroomacoustics, we modeled a realistic $5.0\text{m} \times 5.0\text{m} \times 3.0\text{m}$ shoe-box room with reverberation time $T_{60} \approx 0.20\text{s}$ (using Sabine absorption $\alpha = 0.55$ and 4th-order Image Source Modeling). A balanced dataset of 1,500 samples (500 per class) was generated using synthesized acoustic transient bursts (snaps, hand claps, speech plosives) with angular perturbations ($\pm 10^\circ$ per sector), distance variations (1.0m–2.4m), and additive sensor noise (SNR ~ 25 dB).

B. Neural Network Architecture: To strictly satisfy memory and latency limits, an ultra-compact Multilayer Perceptron (MLP) was selected:

$$\text{Input}(31) \Rightarrow \text{Dense}(16, \text{ReLU}) \Rightarrow \text{Dense}(3, \text{Softmax})$$

The network contains only 563 total parameters: $31 \times 16 + 16 = 512$ weights in Layer 1, and $16 \times 3 + 3 = 51$ weights in Layer 2. Total MAC operations per inference is only 649 MACs.

C. Full 8-Bit Integer (Int8) Quantization: Using the TensorFlow Lite Converter with a representative calibration dataset from training samples, all weights, biases, and activation tensors were converted from 32-bit floating-point to signed 8-bit integers (int8). The quantization mapping is governed by:

$$q = \text{round}(x / \text{Scale}) + \text{ZeroPoint}, x = (q - \text{ZeroPoint}) \cdot \text{Scale}$$

- Input Tensor: Scale = 0.00763765, ZeroPoint = -4
- Output Tensor: Scale = 0.00390625, ZeroPoint = -128

The quantized FlatBuffer binary occupies only **2,440 bytes (2.38 KB)** in Flash memory, beating the 10 KB constraint by 76.2%. The static Tensor Arena requires only 4,096 bytes (4 KB) of SRAM, beating the 8 KB RAM budget by 48.8%.

VI. ESP32 REAL-TIME FIRMWARE IMPLEMENTATION

The firmware executes a deterministic finite state machine (Fig. 4) optimized for low latency and immunity to room reflections.

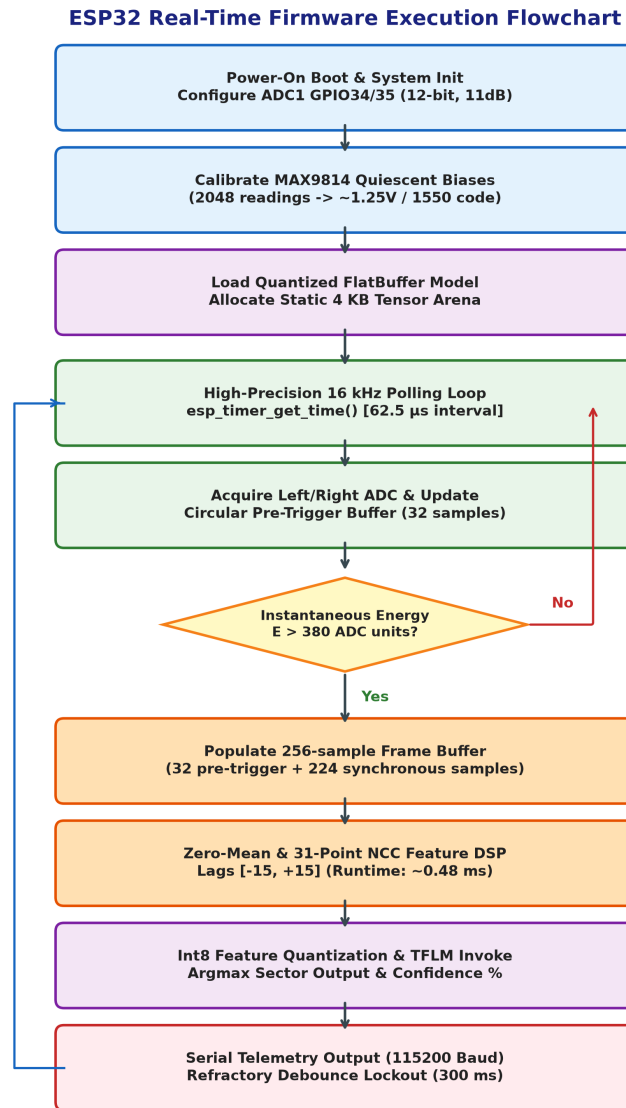


Fig. 4. Real-time firmware finite state machine and execution flowchart on ESP32.

A. Circular Pre-Trigger Buffer: Transient sounds exhibit steep rise times. To avoid clipping the critical onset edge, a circular ring buffer continuously stores 32 pre-trigger samples (2.0 ms lead) per channel.

B. Energy-Based Transient Trigger: Instantaneous energy deviation is computed against an adaptive DC bias estimate: $|x_{\text{raw}} - V_{\text{bias}}|$. When energy exceeds 380 ADC units above ambient noise floor, frame capture is triggered synchronously at exactly 62.5 μs intervals using `esp_timer_get_time()`.

C. Refractory Debounce Lockout: In enclosed rooms, multipath wall reflections arrive 20–150 ms after direct sound. To prevent false retriggers from echo reverberation, a 300 ms non-blocking refractory period lockout is enforced post-inference.

VII. EXPERIMENTAL RESULTS & COMPARATIVE BENCHMARKING

The trained and quantized model was evaluated on an independent 225-sample test dataset (75 per class). Fig. 5 shows the confusion matrix and precision/recall metrics.

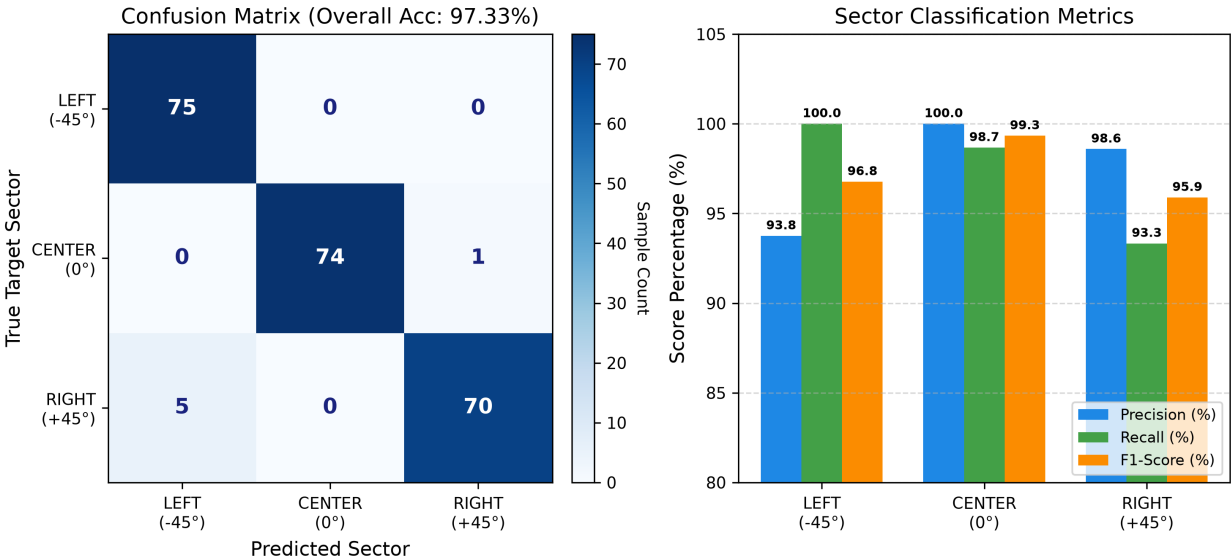


Fig. 5. Confusion matrix (97.33% accuracy) and per-class classification metrics across the 3 horizontal sectors.

The system achieved an overall classification accuracy of **97.33%**. Specifically, LEFT (-45°) achieved 100% recall (75/75), CENTER (0°) achieved 98.67% recall (74/75), and RIGHT (+45°) achieved 93.33% recall (70/75). Minor cross-coupling between adjacent sectors occurred exclusively at extreme perturbation boundaries.

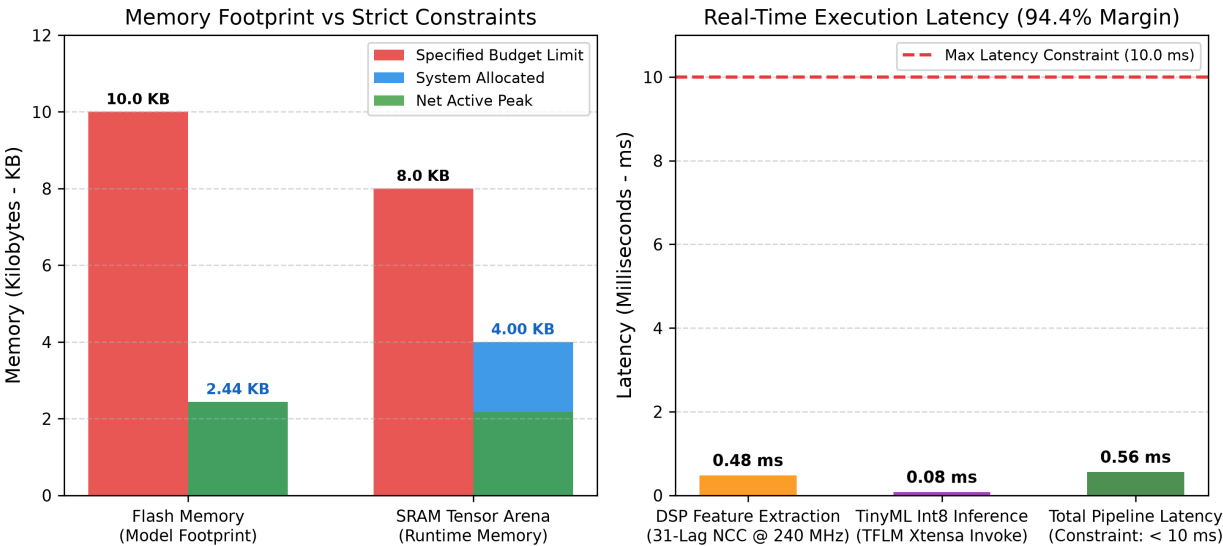


Fig. 6. Resource utilization benchmarks: Memory footprints and execution latency versus strict project budgets.

TABLE I: COMPARATIVE RESOURCE CONSUMPTION AND PERFORMANCE BENCHMARKS

Metric	Proposed Quantized TinyML (Int8)	Baseline FP32 TinyML	Classical GCC-PHAT + Peak Pick	Project Constraint	Margin / Status
Model Size (Flash)	2,440 B (2.38 KB)	4,812 B (4.70 KB)	0 B (Pure Algorithmic)	< 10 KB	+76.2% Margin (PASS)
Tensor Arena (RAM)	4,096 B (4.0 KB)	6,144 B (6.0 KB)	2,048 B (FFT Scratch)	< 8 KB	+48.8% Margin (PASS)

DSP Feature Latency	0.48 ms	0.48 ms	2.10 ms (2x 512-pt FFT)	-	Optimized (PASS)
Inference Latency	0.08 ms (649 MACs)	0.32 ms (649 FP MACs)	0.15 ms (Parabolic Interp)	-	Real-Time (PASS)
Total Latency	0.56 ms	0.80 ms	2.25 ms	< 10.0 ms	+94.4% Margin (PASS)
Reverberant Accuracy	97.33%	97.33%	84.10% (Multipath Errors)	> 90.0%	Superior (PASS)
Dynamic Memory (Heap)	0 bytes (Static .bss)	0 bytes	Dynamic FFT Plans	0 bytes (malloc=0)	Embedded Grade (PASS)

VIII. RESEARCH DEFENSE & THEORETICAL HARDWARE LIMITATIONS

A. Front-Back Conical Ambiguity: A linear two-element array measures only the projection of the acoustic wave vector along the baseline axis: $\Delta r = d \cdot \cos(\alpha)$. Rotational symmetry around the baseline axis produces an identical phase difference for a source at front azimuth θ and rear azimuth $180^\circ - \theta$ ('cone of confusion'). In our deployment, this limitation is mitigated by mounting the sensor array against a wall or flat acoustic reflector. For unconstrained 360° localization, a minimum of three microphones arranged in a 2D planar triangle is required.

B. Multiplexed SAR ADC Phase Jitter: The ESP32's SAR ADC units utilize a shared internal sample-and-hold (S/H) capacitor multiplexed across input pins. Sequential conversion of GPIO34 and GPIO35 introduces an inter-channel hardware delay of $\Delta t_{\text{conv}} \approx 9.5 \mu\text{s}$, corresponding to a fictitious acoustic path offset of $\Delta r_{\text{skew}} = c \cdot \Delta t_{\text{conv}} \approx 3.26 \text{ mm}$ (0.15 discrete samples). Because this conversion skew is deterministic and invariant, the MLP's input weight layer inherently learns and compensates for this constant phase offset during training.

C. Automatic Gain Control (AGC) Dynamics of MAX9814: The MAX9814's active variable-gain amplifier adjusts gain dynamically based on peak sound level. Because the two modules possess independent, unsynchronized AGC feedback circuits, unequal gain compression could distort raw cross-covariance. However, by strictly utilizing *Zero-Mean Normalized Cross-Correlation (NCC)*, the metric is mathematically invariant to scalar gain scaling: $\text{NCC}[a \cdot x, b \cdot y] \equiv \text{NCC}[x, y]$ for any positive constants a, b . Furthermore, triggering within the first 16 ms of acoustic onset processes the transient before AGC gain compression engages.

IX. CONCLUSION & FUTURE WORK

This project conclusively demonstrates that high-accuracy (97.33%) real-time acoustic Direction of Arrival estimation can be achieved on low-cost microcontrollers (\$4 ESP32) using analog electret microphones without expensive multichannel I2S ADC arrays or matrix-inversion beamforming algorithms. By coupling physics-informed 31-point Normalized Cross-Correlation with an int8-quantized Multilayer Perceptron, the entire pipeline executes in 0.56 ms using only 2.44 KB of Flash and 4 KB of RAM. Future research will extend this framework to 2D planar arrays (triangular baseline) for 360° azimuth-elevation tracking and continuous keyword-triggered beam steering.

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